# Automated spike-wave discharge detection in a mouse absence-epilepsy model

*Project specification — event detection on chronic EEG.*

This document is organized in two layers. **Layer 1 — Summary** is a short overview of context, methods, and workflow. **Layer 2 — Reference** mirrors the summary section-for-section and expands each with concrete detail. A terminology glossary sits at the end.

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## Layer 1 — Summary

### 1. Objective

Detect spontaneous spike-wave discharge (SWD) events in chronic EEG recordings from a mouse absence-epilepsy model, and convert those detections into per-session seizure-burden statistics. The system replaces manual visual scoring, which is slow (hours per 24-hour record) and inconsistent between human scorers.

### 2. Setup & data

Freely-moving mice from an absence-epilepsy line, recorded over 24-hour sessions. Two epidural EEG screws (bihemispheric) plus neck EMG, sampled at 256 Hz. The target waveform is a stereotyped spike-and-wave run, roughly 5–7 Hz, lasting about a second or more.

### 3. Ground truth

Labels are generated in-house: trained scorers annotate the lab's own recordings into SWD / non-SWD, ideally with two scorers and adjudication of disagreements. The ground truth is a noisy expert consensus, not a downloaded benchmark.

### 4. Pipeline

Six stages: **preprocess** (filter, artifact handling, normalize) → **window** (short overlapping windows) → **featurize** (spectral, wavelet, morphology, cross-channel) → **model** (gradient-boosted classifier per window) → **post-process** (reconcile overlap, smooth, form events) → **aggregate** (burden statistics).

### 5. Features

Per window: band power in the SWD band and its harmonics, wavelet features, time-domain morphology (line length, amplitude, spike rate), and cross-channel synchrony between the two EEG channels. Assembled into one row per window.

### 6. Model & training

CatBoost (gradient boosting) on the tabular per-window features, trained against expert labels. Validation splits by animal and by time, and reports agreement-based metrics rather

than raw accuracy. A deep-learning variant replaces stages 3–5 with a 1-D network on the raw windowed signal.

### 7. Post-processing & event formation

Overlapping window predictions are reconciled into a time-resolved decision, thresholded with hysteresis, smoothed, then grouped into discrete events (with merge-gap and minimum-duration rules) so each true event is counted exactly once.

### 8. Validation & metrics

Event-level precision/recall against annotated events (with an overlap tolerance), Cohen's κ against scorers, false-alarm rate, and detection lead/latency — computed only after event formation.

### 9. Outputs

A per-session event list (onset, offset, duration) and derived statistics: event count, events per hour, total time in SWD (seizure burden), duration distribution, circadian patterning, and cross-condition comparisons.

### 10. Scope & variants

Core scope is **offline** detection with bidirectional smoothing. A **real-time causal** variant (persistence-counter logic, latency budget) is the relevant form for responsive-neurostimulation contexts. Feature-based (CatBoost) and deep-learning (1-D CNN/TCN) are interchangeable modeling backbones.

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## Layer 2 — Reference

### 1. Objective

The absence-epilepsy phenotype is defined by recurrent SWDs — brief, stereotyped, generalized electrographic events. Quantifying them across long recordings is the core readout for genotype, drug, and condition studies, expressed as *seizure burden* (how often and how long SWDs occur).

Manual scoring is the status quo and the bottleneck: a skilled human needs on the order of hours to score a single 24-hour record, which pushes labs toward scoring only representative snippets, and visual scoring shows meaningful disagreement between scorers. The goal is an automated detector that (a) reproduces expert scoring at expert-level agreement, (b) scales to full continuous records, and (c) produces reproducible burden statistics.

The task is framed as **binary event detection** on a continuous multichannel signal: locate every SWD event in time.

## 2. Setup & data

**Preparation.** Freely-moving mice from an absence-epilepsy line whose hallmark is spontaneous SWDs. Recordings are chronic and continuous, typically in 24-hour sessions to capture circadian structure.

**Montage.**

- 2 EEG channels — epidural stainless-steel screws in contact with the dura, one over each cortical hemisphere.
- 1 reference screw and 1 ground screw (commonly posterior / over cerebellum).
- 2 EMG wires in the neck muscle, for movement/state context and artifact rejection.

**Sampling.** 256 Hz is a sensible floor (many labs run 400–500 Hz). The SWD fundamental is low (~5–7 Hz in this model; some absence models/descriptions cite 3–5 Hz), but the sharp spike component carries energy well above the fundamental, so the rate must preserve spike morphology.

**Why only two EEG channels.** Two channels is the standard for rodent seizure/sleep work, for three reasons: the mouse skull physically limits electrode count; generalized SWDs appear bilaterally and near-synchronously, so two sites are sufficient to capture them; and the *relationship between* the two channels (synchrony) becomes a discriminative feature that a single channel cannot provide. Higher-density mouse arrays exist (16–32 channels) but are not required here.

**Signal target.** A stereotyped spike-and-wave run: repetitive high-amplitude spike-and-slow-wave complexes, bilaterally synchronous, ~5–7 Hz, ~1 s or longer, with abrupt onset and offset against a lower-amplitude background.

## 3. Ground truth & annotation

**Source.** Labels are produced in-house by trained scorers annotating the lab's own recordings — not obtained from a public corpus. (Public expert-annotated EEG corpora are abundant but overwhelmingly human-clinical; they are useful as methodology references, but the target labels for this mouse montage come from the lab.)

**Protocol.** Scorers mark SWD event onsets and offsets. For robust ground truth, two independent scorers annotate and a third adjudicates disagreements. This yields a consensus label set with a quantified inter-scorer agreement.

**Label characterization.** The ground truth is treated as a *noisy consensus*, not an oracle. Design consequences: report model performance against multiple scorers where available; consider confidence-weighted or consensus targets; and interpret the ceiling of model performance relative to human–human agreement rather than a hypothetical perfect label.

**Window labeling.** Because the model operates on fixed windows (stage 4) while annotations are continuous event spans, each window inherits a label from its overlap with

annotated events (e.g., positive if it overlaps an SWD event by a set fraction).

## 4. Processing pipeline

The pipeline has six stages. Stages 1–5 turn continuous signal into per-window predictions; stage 6 turns predictions into events and statistics.

### 4.1 Preprocess

- **Filtering:** band-pass to the physiological range of interest; notch at line frequency (50/60 Hz).
- **Re-referencing:** against the reference screw; optionally derive bipolar channels.
- **Normalization:** per-channel, per-session z-scoring so amplitude drift across animals and days does not leak into features.
- **Artifact handling:** the dominant real-world problem. A freely-moving mouse produces movement and EMG artifact (chewing, grooming) that can mimic high-amplitude spikes and is the principal false-positive source. EMG channels and amplitude/impedance heuristics help gate these epochs.

### 4.2 Windowing

- **Window length:** ~1 s (256 samples per channel at 256 Hz). Short, because SWDs are brief and a long window dilutes a short event in mostly-baseline signal.
- **Stride (hop):** ~0.5 s (50% overlap), i.e., a new window every half second. Overlap sharpens onset/offset localization, multiplies training examples, and prevents an event from being split across a boundary.
- **Output:** a sequence of `[channels × samples]` windows, each of which becomes one featurized record.

Window length and stride are the two decomposition parameters: length sets how much signal each record sees (and its spectral resolution); stride sets how often a record is emitted (temporal resolution / detection cadence).

### 4.3 Featurize

Per window, compute:

- **Spectral:** power spectral density / band power in the SWD fundamental band and its harmonics.
- **Wavelet:** wavelet-decomposition features targeting the SWD frequency and harmonics (captures the transient spike-and-wave structure better than a plain FFT).
- **Time-domain morphology:** line length, amplitude/variance, spike rate/count — encoding the spike-and-wave shape.
- **Cross-channel:** synchrony / coherence between the two EEG channels, exploiting the bilateral synchrony of generalized SWDs.

These stack into a tabular dataset: one row per window, feature columns, one label. This is the representation consumed by standard tabular models.

### 4.4 Model

- **Backbone:** CatBoost (gradient-boosted trees) on the per-window feature table, predicting SWD / non-SWD.
- **Alternative (deep-learning variant):** a 1-D CNN or temporal model on the raw windowed signal, which learns its own features and collapses stages 3–5 into one block. Same windowing front-end; no explicit featurization or tabular intermediate.

### 4.5 Post-process → 4.6 Aggregate

Covered in §7 below (kept as one block there because the sub-steps are tightly coupled).

## 5. Features (detail)

| Group | Features | Rationale |
|---|---|---|
| Spectral | band power in SWD band + harmonics | SWDs concentrate energy at a characteristic fundamental and harmonics |
| Wavelet | wavelet coefficients / band energies at SWD scales | resolves the transient spike-and-wave morphology |
| Morphology | line length, amplitude, variance, spike rate | captures spike-and-wave shape in the time domain |
| Cross-channel | inter-channel synchrony / coherence | generalized SWDs fire near-synchronously across hemispheres |

The cross-channel group is the one feature family that a single-channel montage cannot produce, and is often among the most discriminative for generalized events.

## 6. Model & training (detail)

**Targets.** Binary per-window label derived from expert event annotations by overlap.

**Splitting.** Split by **animal** (never mix windows from one mouse across train/test) and preferably **time-ordered** (train on earlier sessions, test on later) so performance estimates reflect future recordings rather than interpolation within known ones. Windows from the same recording are highly correlated; splitting by window inflates apparent performance.

**Class imbalance.** SWD windows are a minority of all windows; handle with class weighting, threshold tuning, and imbalance-aware metrics rather than raw accuracy.

**Model comparison.** CatBoost/gradient boosting is the feature-based baseline and typically strong on tabular features. The deep-learning variant is the upgrade path where learned

features on the raw signal can exceed hand-crafted ones; it trades interpretability and data-efficiency for representational power.

## 7. Post-processing & event formation

Raw per-window predictions are not an event list — with overlap, one true event produces a *run* of consecutive positive windows, plus scattered single-window blips. Five coupled steps resolve this:

1. **Reconcile overlap.** Each time point is covered by multiple overlapping windows; average their predictions (overlap-add) into a per-sample probability trace — turning redundant votes into a single robust decision rather than double counts.
2. **Threshold with hysteresis.** Binarize with a dual threshold (higher to *enter* the event state, lower to *stay*) to prevent flicker at event boundaries.
3. **Smooth.** Median filter / majority vote / HMM over the prediction sequence to enforce temporal coherence.
4. **Form events.** Run-length-group consecutive positives into discrete events with onset/offset. Apply a **merge-gap** rule (fuse events separated by a sub-threshold gap, so a brief dropout does not split one event into two) and a **minimum-duration** rule (discard events shorter than the physiological floor, killing isolated false positives). After this step, each true event is counted exactly once.
5. **Aggregate.** From the clean event list, compute session statistics (see §9).

Event formation (step 4) is the deduplication stage and is a prerequisite for honest event-level evaluation (§8) — you cannot score events until events exist.

## 8. Validation & metrics

Metrics are computed at the **event level**, after event formation:

- **Event-level precision / recall (and F1):** a detected event is a true positive if it overlaps an annotated event within a tolerance collar. This is what the science cares about, and it differs from window-level scoring (which rewards overlap redundancy and overstates performance).
- **Cohen's κ** against expert scorers, interpreted relative to human–human agreement.
- **False-alarm rate** (false events per hour) — the metric that most governs operational trust.
- **Detection lead / latency** — how early or late detected onsets are relative to annotated onsets (critical for any causal/real-time variant).

Window-level metrics (per-window F1) may be reported as an intermediate diagnostic but are not the headline.

### 9. Outputs & deliverables

- **Per-session event list:** each SWD with onset, offset, duration.
- **Seizure burden:** event count, events per hour, total time in SWD.
- **Duration distribution:** mean and spread of event lengths.
- **Circadian structure:** how SWD occurrence varies across the light/dark cycle.
- **Comparisons:** burden differences across genotype, drug/vehicle, or other conditions — the actual scientific readout.

The deliverable is the burden statistics, not per-window labels; producing correct burden numbers is the difference between a model and a usable tool.

### 10. Scope, variants & extensions

**Offline vs real-time.** The core scope is offline analysis, which may smooth **bidirectionally** (using signal before *and* after a candidate event to fix its boundaries). A **real-time causal** variant may only use past signal; the analog of the post-processing smoothing becomes a **persistence counter** (require N consecutive positive windows before declaring an event), trading detection latency for fewer false triggers. This causal form is the one relevant to responsive-neurostimulation / closed-loop contexts, where stride corresponds to the control-cycle cadence and latency is a hard constraint.

**Feature-based vs deep-learning.** The two modeling backbones (§4.4) are interchangeable at the model stage and share the same windowing front-end, post-processing, and evaluation. The deep variant is the natural upgrade when hand-crafted features become the performance ceiling.

**Multichannel extension.** A higher-density mouse array would add spatial features (topography, propagation) but is not required for generalized SWD detection and adds surgical and preprocessing burden.

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## Terminology

- `SWD` — **Spike-Wave Discharge.** The stereotyped, generalized spike-and-slow-wave electrographic event that defines absence seizures; the detection target here.
- `absence epilepsy` — A generalized epilepsy characterized by brief lapses of awareness with SWDs on EEG; modeled in specific mouse lines with spontaneous SWDs.
- `epidural EEG` — EEG recorded from screw electrodes contacting the dura (outside the brain), the standard chronic rodent montage.
- `EMG` — **Electromyography.** Muscle-activity signal (here from neck wires), used for behavioral-state context and artifact rejection.
- `epoch / window` — A fixed-length slice of the continuous signal that is featurized and labeled as one record. (In sleep contexts "epoch" is standard; here "window" is

used to avoid confusion with training epochs.)

- `stride / hop` — The step between consecutive windows. Stride < window length ⇒ overlapping windows.
- `band power` — Signal power within a frequency band; a spectral feature.
- `line length` — Sum of absolute sample-to-sample differences over a window; a time-domain morphology feature sensitive to spiky, high-frequency content.
- `coherence / synchrony` — Measures of how consistently two channels co-vary; high across hemispheres during generalized SWDs.
- `hysteresis (dual threshold)` — Using a higher threshold to enter a detected state and a lower one to remain, to suppress boundary flicker.
- `merge-gap` — A post-processing rule fusing two detected events separated by a gap shorter than a set duration.
- `minimum duration` — A post-processing rule discarding detected events shorter than a physiological floor.
- `event-level scoring` — Evaluation that credits a detection when it overlaps an annotated event within a tolerance, as opposed to per-window scoring.
- `tolerance collar` — The allowed timing slack around an annotated event within which a detection still counts as correct.
- `Cohen's κ` — An agreement metric corrected for chance; used against expert scorers and between scorers.
- `seizure burden` — Aggregate SWD load per session (event count, events/hour, total time in SWD).
- `persistence counter` — A causal detection rule requiring N consecutive positive windows before declaring an event; the real-time analog of smoothing.